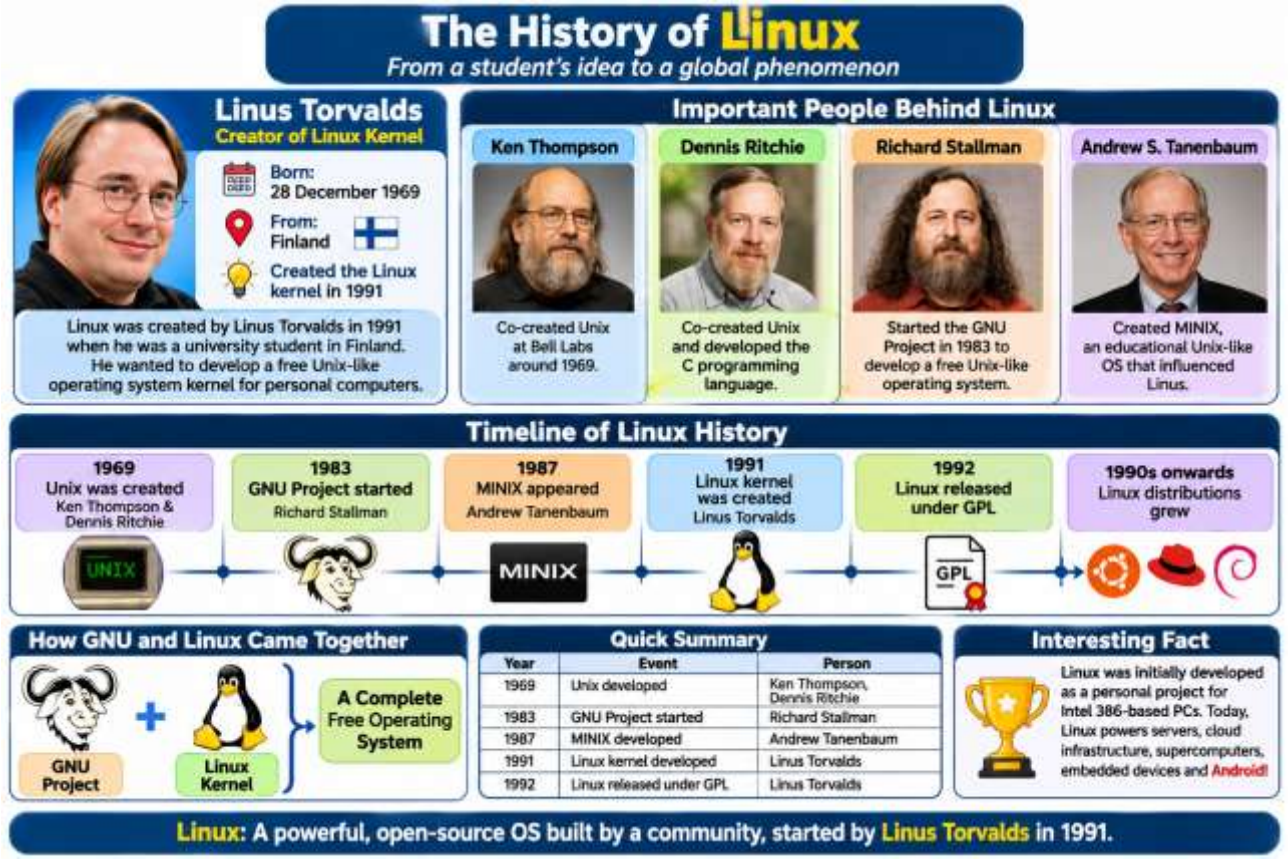
Linux

Linux is an open-source operating system (OS) that manages computer hardware and software and allows users to run applications, execute commands, manage files, and control system resources.

Linux was created by **Linus Torvalds in 1991** when he was a university student in Finland. He wanted to develop a free Unix-like operating system kernel for personal computers.

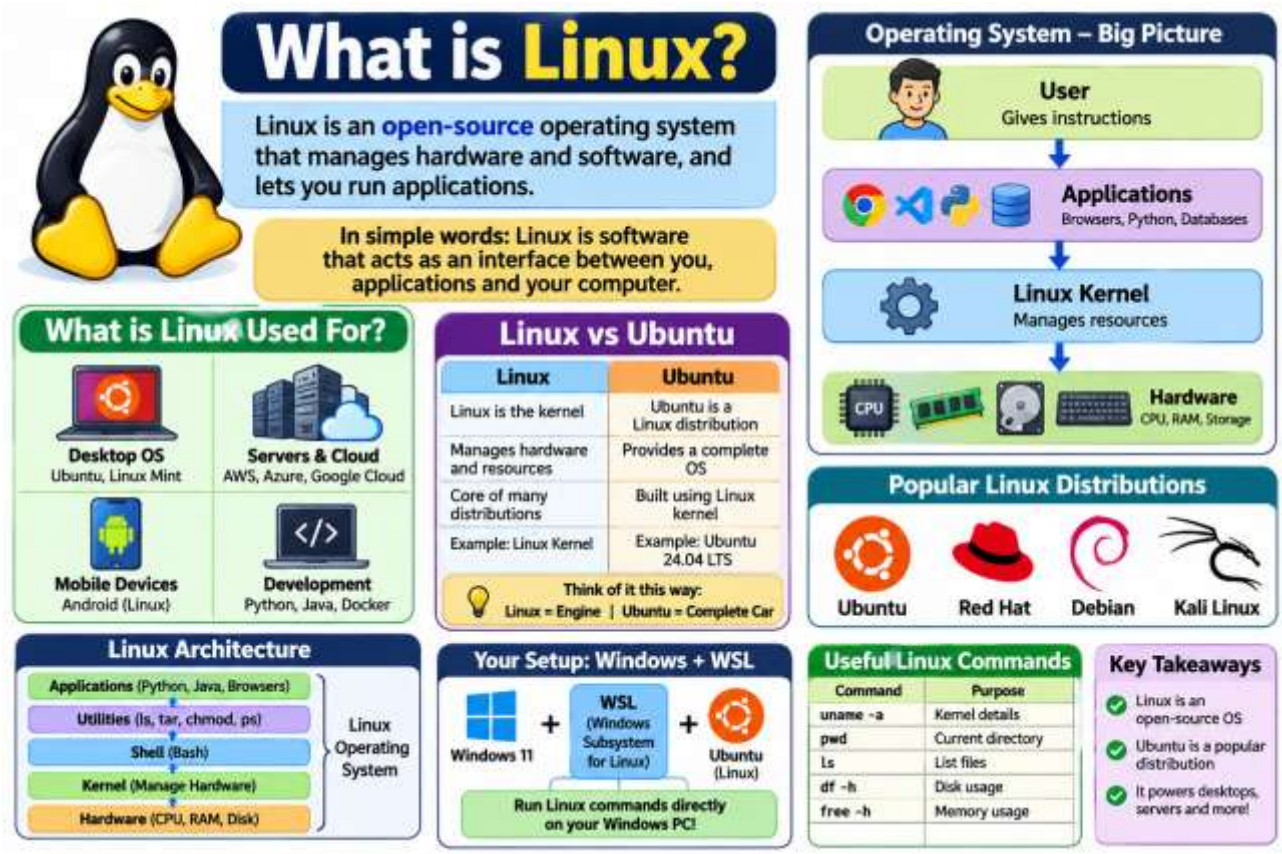


In simple words, Linux is software that acts as an interface between the user, applications, and computer hardware.

We can think of Linux as an alternative to Microsoft Windows.

1. What does an operating system do?

An operating system manages the computer's resources



2. What is the difference between Linux and Ubuntu?

This is an important distinction.

Linux	Ubuntu
Linux is the kernel	Ubuntu is a Linux distribution
It manages hardware and system resources	It provides a complete operating system
Core of many Linux distributions	Built using the Linux kernel
Example: Linux kernel	Example: Ubuntu 24.04 LTS

Think of it this way:

- **Linux = Engine**
- **Ubuntu = Complete car using that engine**

Other popular Linux distributions include:



Ubuntu



Red Hat



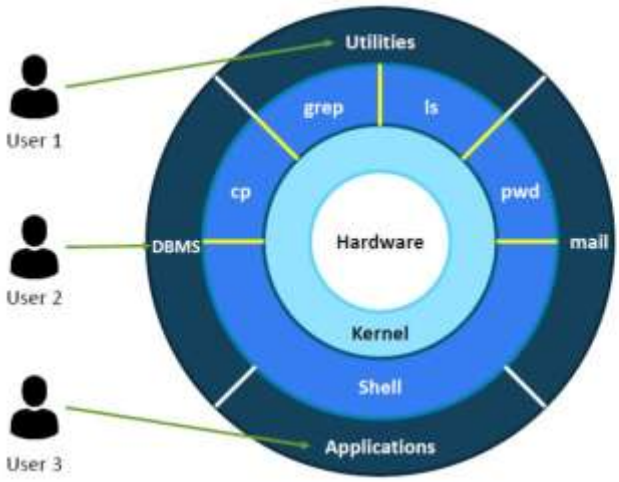
Debian

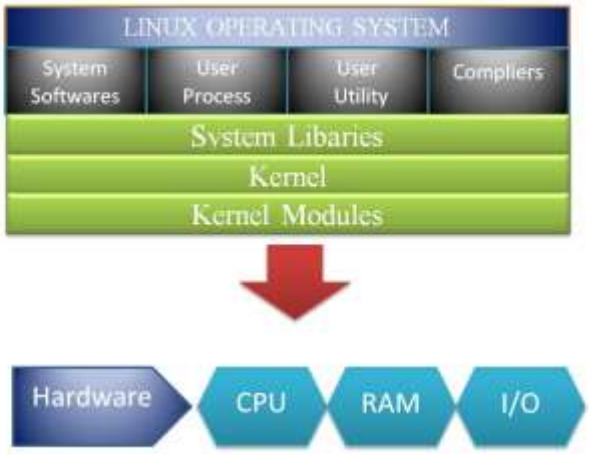


Kali Linux

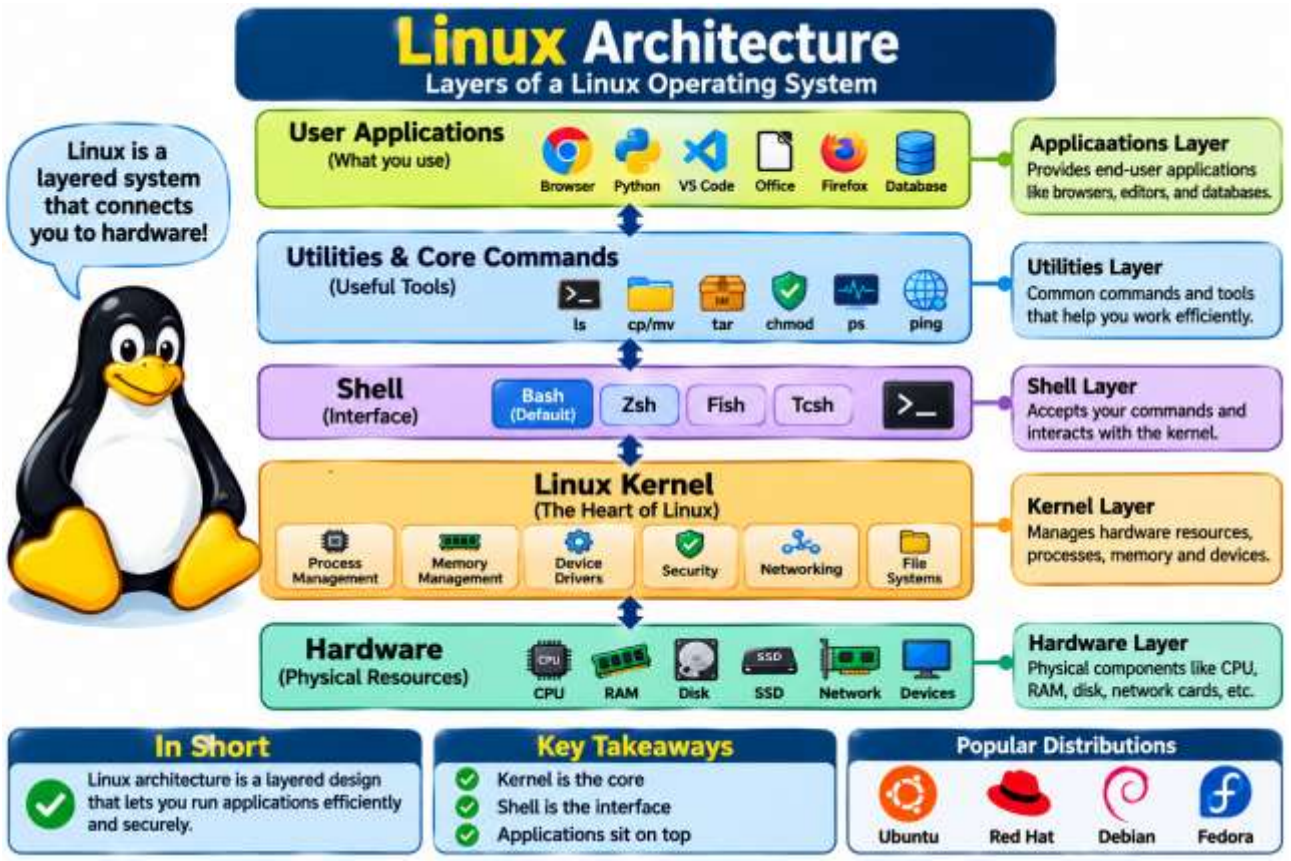
4. Linux architecture

Linux can be understood through these layers:





Layer	Purpose
Hardware	CPU, RAM, disk
Kernel	Manages hardware, memory, processes and devices
Shell	Accepts and interprets commands
Utilities	Commands such as `ls`, `chmod`, `tar`, `ps`
Applications	Python, Java, browsers, databases

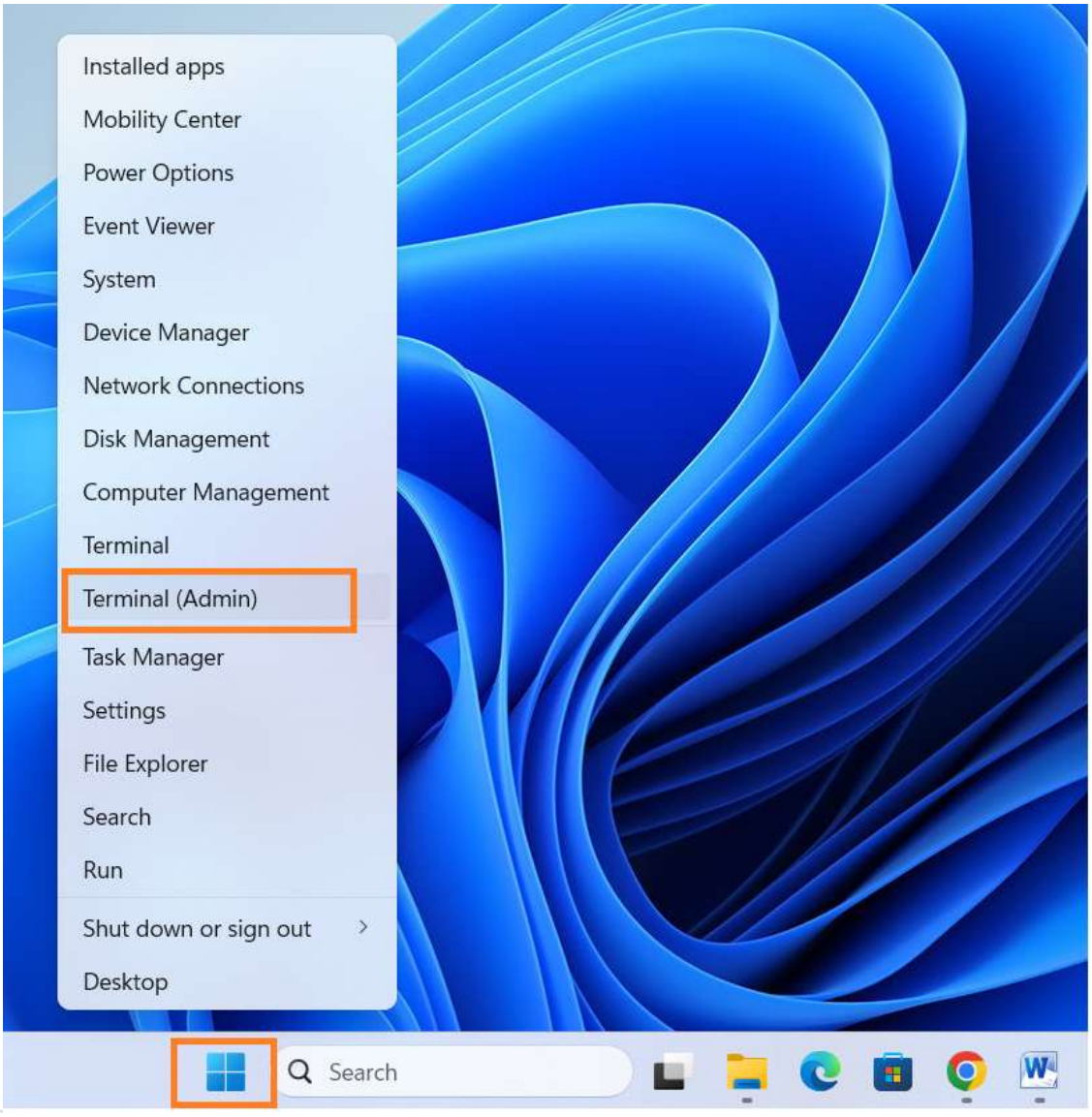


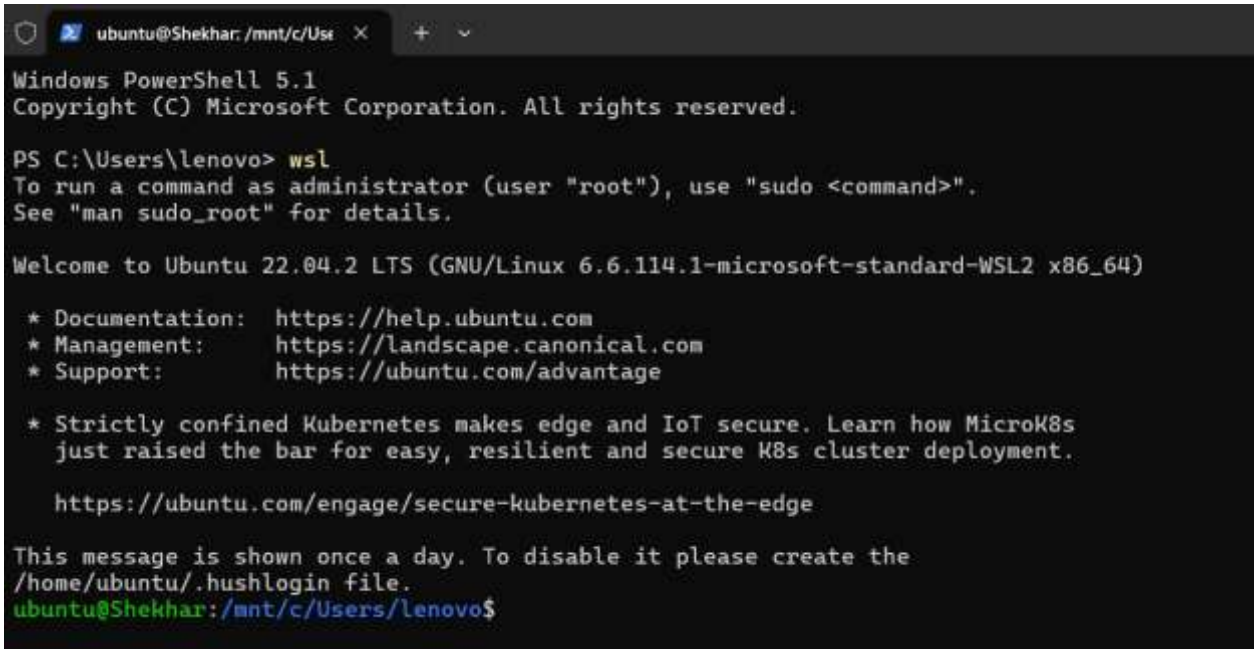
Step-by-Step Practical Guide for Ubuntu on Windows 11 (WSL)

Learning objective: Learn how to navigate Linux, manage files and directories, search content, manage users and permissions, monitor processes and system resources, use networking tools, install packages, and create archives.

Important environment note: Run Linux commands inside the Ubuntu terminal. If the prompt begins with PS C:\, you are in Windows PowerShell.

Open Terminsl(Admin):





PS C:\Users\lenovo> wsl

To install Ubuntu (run PowerShell as Administrator):

wsl --install -d Ubuntu

After installation, restart if requested, open Ubuntu, and create a Linux username and password. Password characters do not appear while typing; this is normal.

1. Open and Identify the Linux Terminal

Use Ubuntu through Windows Subsystem for Linux (WSL). Once Ubuntu opens, confirm that you are in Linux before running the commands in this guide.

```
pwd
whoami
uname -a
```

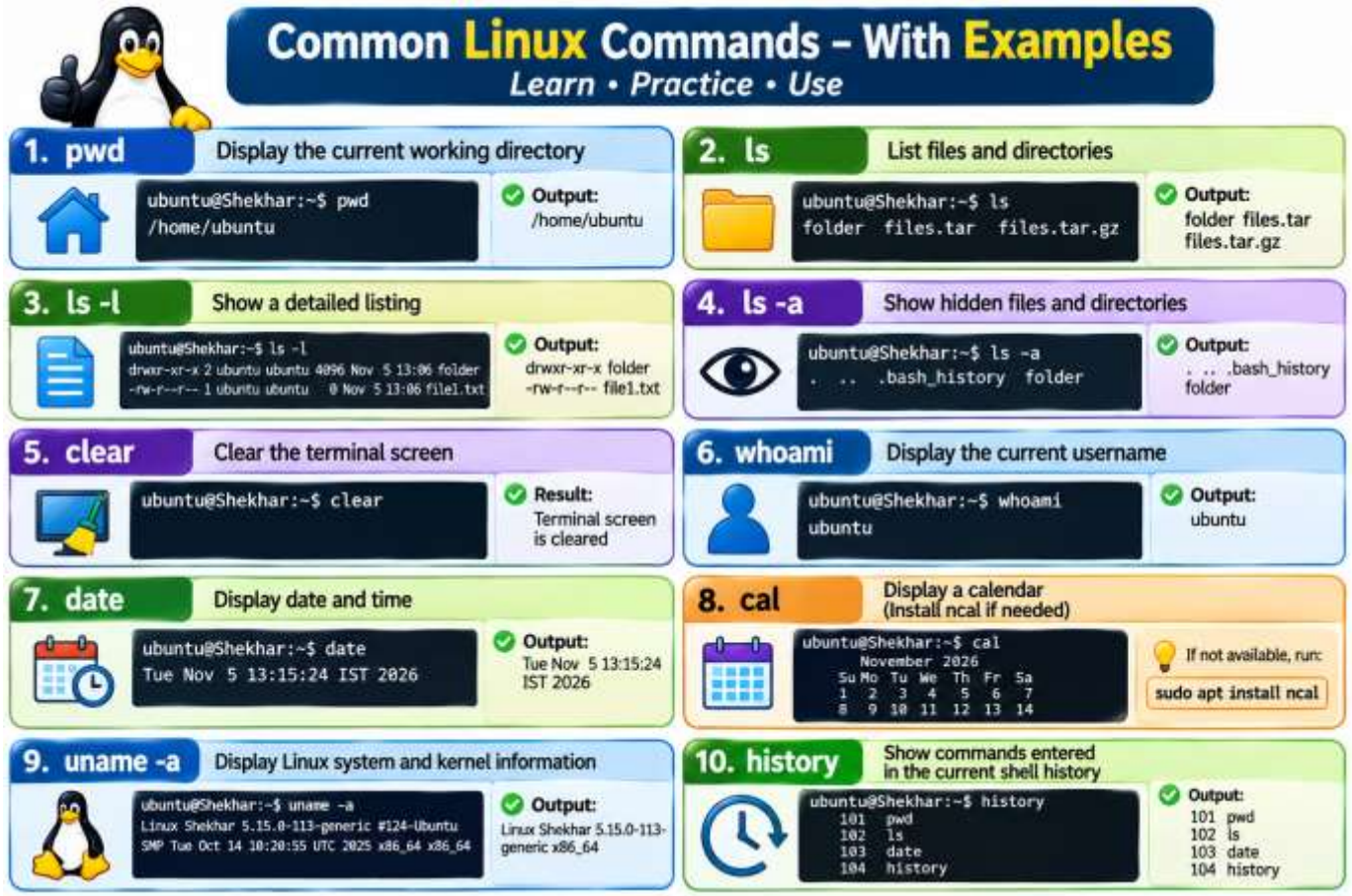
Access Windows files from Ubuntu: Windows drives are usually mounted **under /mnt.** For example, the Windows user folder shown in the earlier terminal session can be reached with:

```
cd /mnt/c/Users/cc pwd
ls -la
```

Do not run cd /mnt/c/Users/Bccat a PowerShell prompt. It is a Linux path and must be entered inside Ubuntu.

2. Basic Linux Commands

No.	Command	Purpose
1	pwd	Display the current working directory
2	ls	List files and directories
3	ls -l	Show a detailed listing
4	ls -a	Show hidden files and directories
5	clear	Clear the terminal screen
6	whoami	Display the current username
7	date	Display date and time
8	cal	Display a calendar (install ncal if unavailable) **sudo apt install ncal**
9	uname -a	Display Linux system and kernel information
10	history	Show commands entered in the current shell history

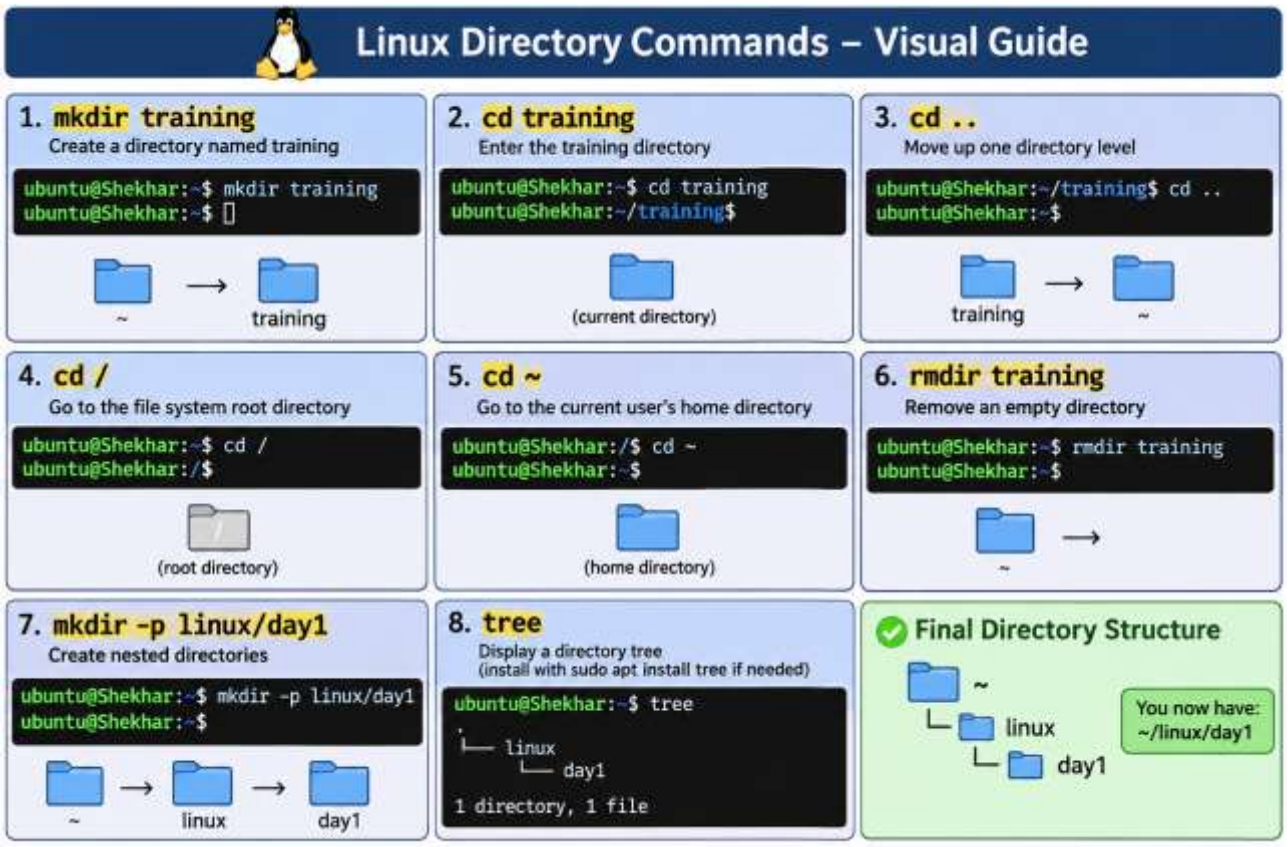

Common Linux Commands – With Examples
Learn • Practice • Use
1. pwd
Display the current working directory
ubuntu@Shekhar:~$ pwd
/home/ubuntu
Output:
/home/ubuntu
2. ls
List files and directories
ubuntu@Shekhar:~$ ls
folder files.tar files.tar.gz
Output:
folder files.tar
files.tar.gz
3. ls -l
Show a detailed listing
ubuntu@Shekhar:~$ ls -l
drwxr-xr-x 2 ubuntu ubuntu 4096 Nov 5 13:06 folder
-rw-r--r-- 1 ubuntu ubuntu 0 Nov 5 13:06 file1.txt
Output:
drwxr-xr-x folder
-rw-r--r-- file1.txt
4. ls -a
Show hidden files and directories
ubuntu@Shekhar:~$ ls -a
. .. .bash_history folder
Output:
. .. .bash_history
folder
5. clear
Clear the terminal screen
ubuntu@Shekhar:~$ clear
Result:
Terminal screen
is cleared
6. whoami
Display the current username
ubuntu@Shekhar:~$ whoami
ubuntu
Output:
ubuntu
7. date
Display date and time
ubuntu@Shekhar:~$ date
Tue Nov 5 13:15:24 IST 2026
Output:
Tue Nov 5 13:15:24
IST 2026
8. cal
Display a calendar
(Install ncal if needed)
ubuntu@Shekhar:~$ cal
November 2026
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
If not available, run:
sudo apt install ncal
9. uname -a
Display Linux system and kernel information
ubuntu@Shekhar:~$ uname -a
Linux Shekhar 5.15.0-113-generic #124-Ubuntu
SMP Tue Oct 14 10:20:55 UTC 2025 x86_64 x86_64
Output:
Linux Shekhar 5.15.0-113-
generic x86_64
10. history
Show commands entered
in the current shell history
ubuntu@Shekhar:~$ history
101 pwd
102 ls
103 date
104 history
Output:
101 pwd
102 ls
103 date
104 history

3. Directory Management

Command	Purpose
mkdir training	Create a directory named training
cd training	Enter the training directory
cd ..	Move up one directory level
cd /	Go to the file system root directory
cd ~	Go to the current user's home directory ubuntu@Shekhar:**~$ cd /mnt/c/Users/lenovo**
rmdir training	Remove an empty directory
mkdir -p linux/day1	Create nested directories
tree	Display a directory tree (install with sudo apt install tree if needed)

sudo snap install tree # version 2.1.3+pkg-5852, or

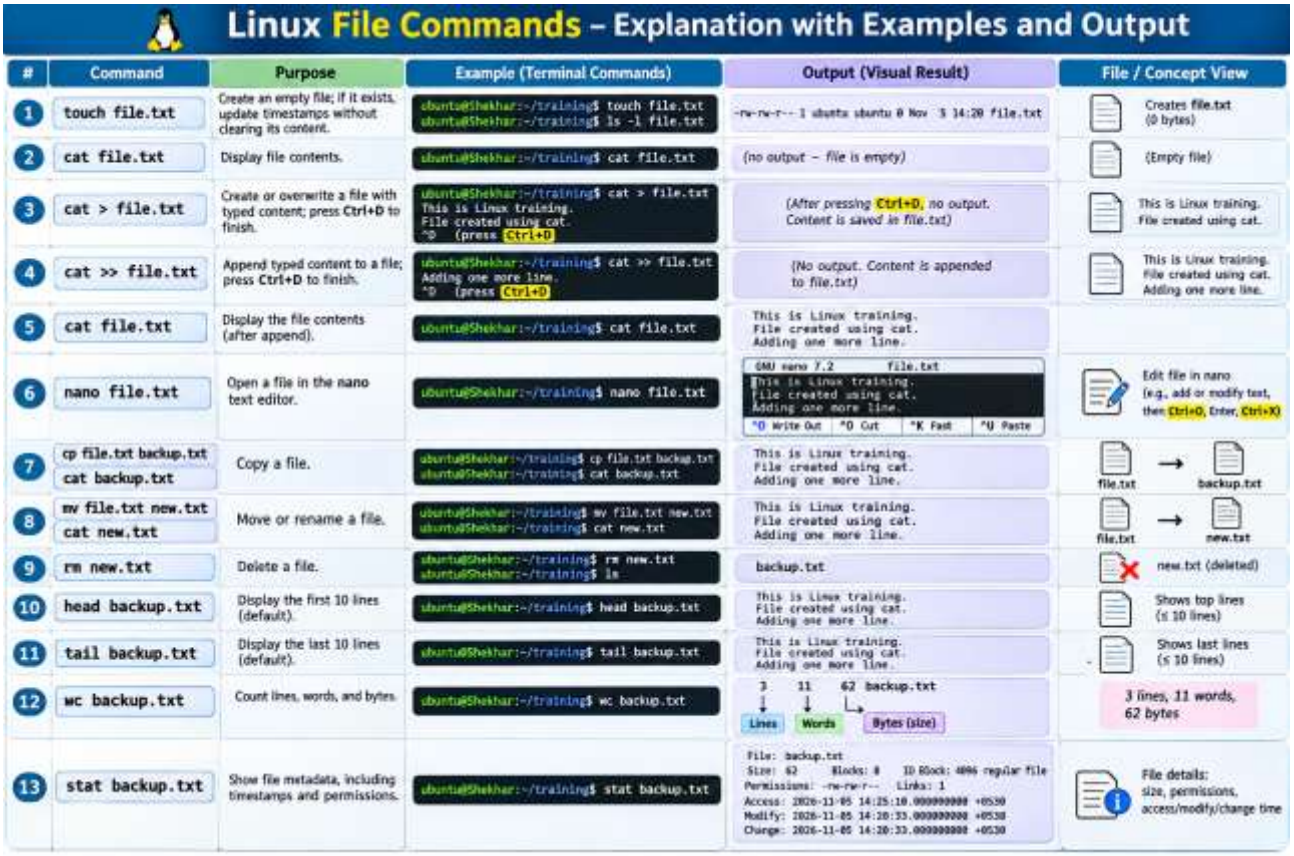
sudo apt install tree # version 2.0.2-1



Note: **rmdir** removes only empty directories. Use **rm -r** for non-empty directories only when you are certain you want to delete their contents.

4. File Management Commands

Command	Purpose
touch file.txt	**Create an empty file**; if it exists, update timestamps without clearing its content
cat file.txt	Display file contents
cat > file.txt	**Create or overwrite** a file with **typed content**; press **Ctrl+D** to finish
cat >> file.txt	Append typed content to a file; press **Ctrl+D** to finish-- **cat file.txt**
nano file.txt	Open a file in the nano text editor
cp file.txt backup.txt **cat backup.txt**	Copy a fil
mv file.txt new.txt **cat new.txt**	Move or rename a file
rm new.txt	Delete a file
head backup.txt	Display the first 10 lines
tail backup.txt	Display the last 10 lines
wc backup.txt	Count lines, words and bytes
file backup.txt	Identify the file type
stat backup.txt	Show file metadata, including timestamps and permissions



Now write following commands for step 5

ubuntu@Shekhar:/mnt/c/Users/lenovo$ touch students.txt

ubuntu@Shekhar:/mnt/c/Users/lenovo$ cat > students.txt

Rahul

Vijay

Ajay

ubuntu@Shekhar:/mnt/c/Users/lenovo$

5. File Searching Commands

Command	Purpose
find . -name "*.txt"	Find .txt files under the current directory
grep "Rahul" students.txt	Find lines containing Rahul
grep -i "rahul" students.txt	Search without case sensitivity
grep -n "Rahul" students.txt	Show matching line numbers
locate students.txt	Search using the locate database (install mlocate/plocate and update its database if needed)-- **sudo apt install plocate**
which python3	Show the executable path for python3, if installed

Linux File Searching Commands – Hands-on Example & Outputs

1. Create and Add Data to File (Setup)

```
ubuntu@Shekhar:/mnt/c/Users/lenovo$ touch students.txt
ubuntu@Shekhar:/mnt/c/Users/lenovo$ cat > Students.txt
Rahul
Vijay
Ajay
ubuntu@Shekhar:/mnt/c/Users/lenovo$
```

(Type the lines and press **Ctrl + D** to save)

File Created: Students.txt



Students.txt
Rahul
Vijay
Ajay

File Searching Commands

#	Command	Purpose	Example & Output (Terminal)	Visual Explanation

1	`find . -name "*.txt"`	Find .txt files under the current directory (recursively).	`ubuntu@Shekhar:/mnt/c/Users/lenovo$ find . -name "*.txt"`
			`./students.txt`
			`./Students.txt`
				• Searches from current folder (.)
				• Lists all .txt files.

2	`grep "Rahul" students.txt`	Find lines containing exact text "Rahul" (case-sensitive).	`ubuntu@Shekhar:/mnt/c/Users/lenovo$ grep "Rahul" students.txt` `Rahul`	• Matches only "Rahul" (with same case).
3	`grep -i "rahul" students.txt`	Search without case sensitivity ("Rahul", "rahul", "RAHUL").	`ubuntu@Shekhar:/mnt/c/Users/lenovo$ grep -i "rahul" students.txt` `Rahul`	• -i = ignore case (matches any case).
4	`grep -n "Rahul" students.txt`	Show matching line numbers.	`ubuntu@Shekhar:/mnt/c/Users/lenovo$ grep -n "Rahul" students.txt` `1:Rahul`	• Shows the line number (1) where "Rahul" is found.

5	`locate students.txt`	Search using the locate database (install plocate and update its database if needed).	`ubuntu@Shekhar:~$ sudo apt install plocate`
			`... (installs the locate database tool)`
			`ubuntu@Shekhar:~$ sudo updatedb`
			`ubuntu@Shekhar:~$ locate students.txt`
			`/mnt/c/Users/lenovo/students.txt`
			`/mnt/c/Users/lenovo/Students.txt`
				• Uses a pre-built index (database) for fast search.
				• Run `sudo updatedb` after installing to refresh the index.

6	`which python3`	Show the executable path for python3, if installed.	`ubuntu@Shekhar:~$ which python3` `/usr/bin/python3`	• Shows the full path of the python3 executable.

6. Linux User Management

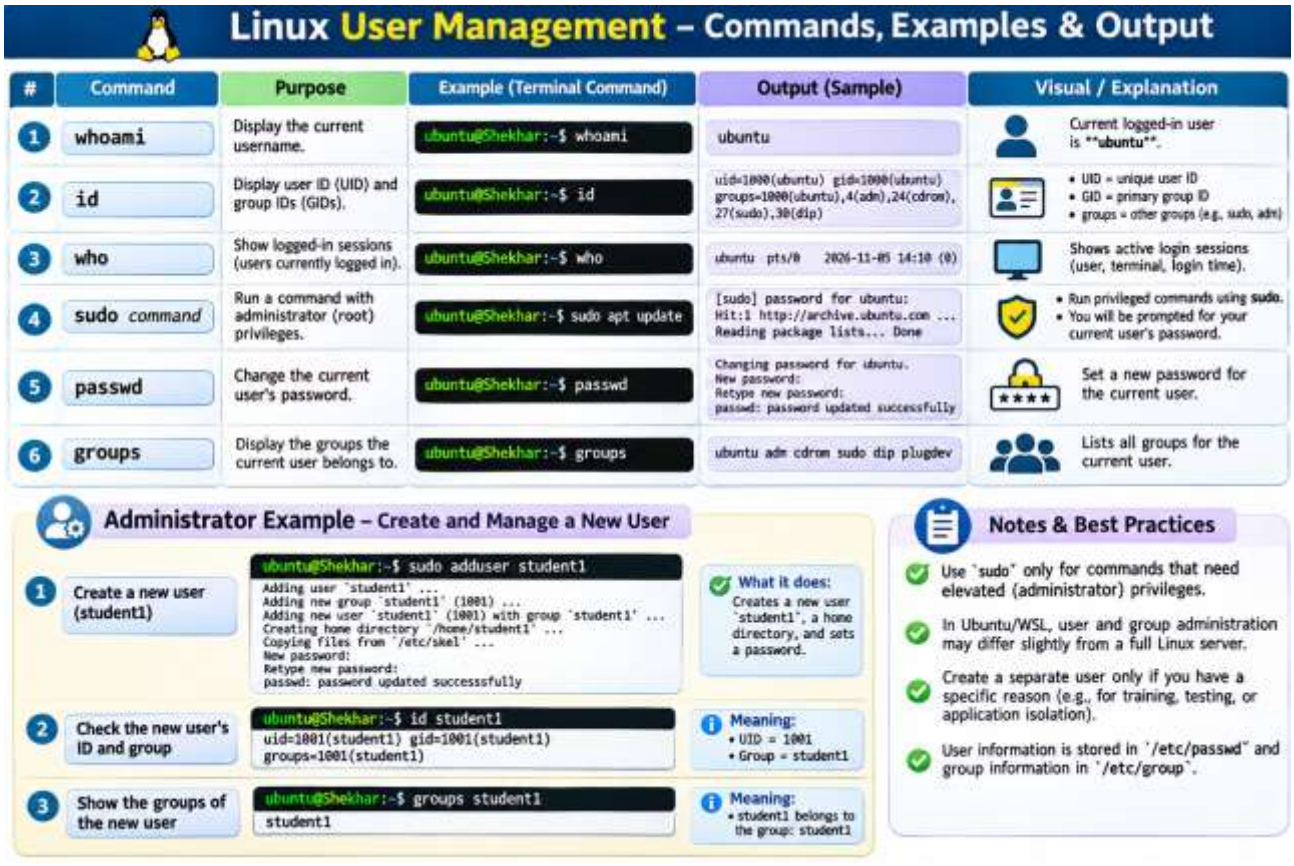
Command	Purpose
whoami	Display the current username
id	Display user ID and group IDs
who	Show logged-in sessions
sudo command	Run a command with administrator privileges
passwd	Change the current user's password
groups	Display the groups the current user belongs to

Administrator example: Create a separate user only if you have a reason to do so. You may be asked for your current user's password.

```
sudo adduser
student1 id
student1
groups student1
```

In Ubuntu/WSL, user and group administration may differ slightly from a full Linux server.

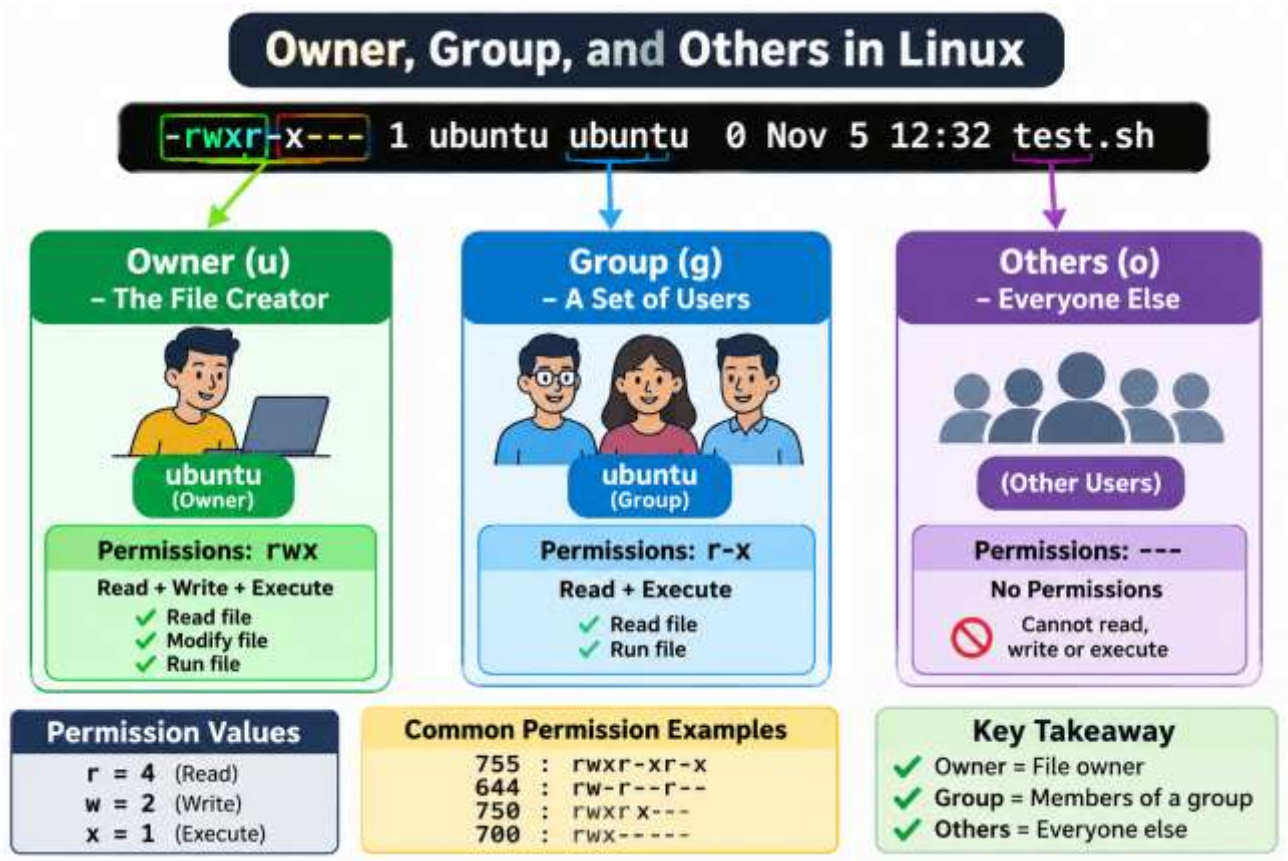
Use sudo only for commands that require elevated privileges.



7. Linux File Permissions

Linux permissions determine who can read, write, or execute a file. The three permission groups are owner, group, and others.

Permission	Numeric value	Meaning
Read (r)	4	Read file contents
Write (w)	2	Modify file contents
Execute (x)	1	Run a file as a program/script



Example: Create a script file and inspect or change its permissions.

ubuntu@Shekhar:/mnt/ Users/lenovo$ **cd ~**

ubuntu@Shekhar:~$ **mkdir PermissionLab**

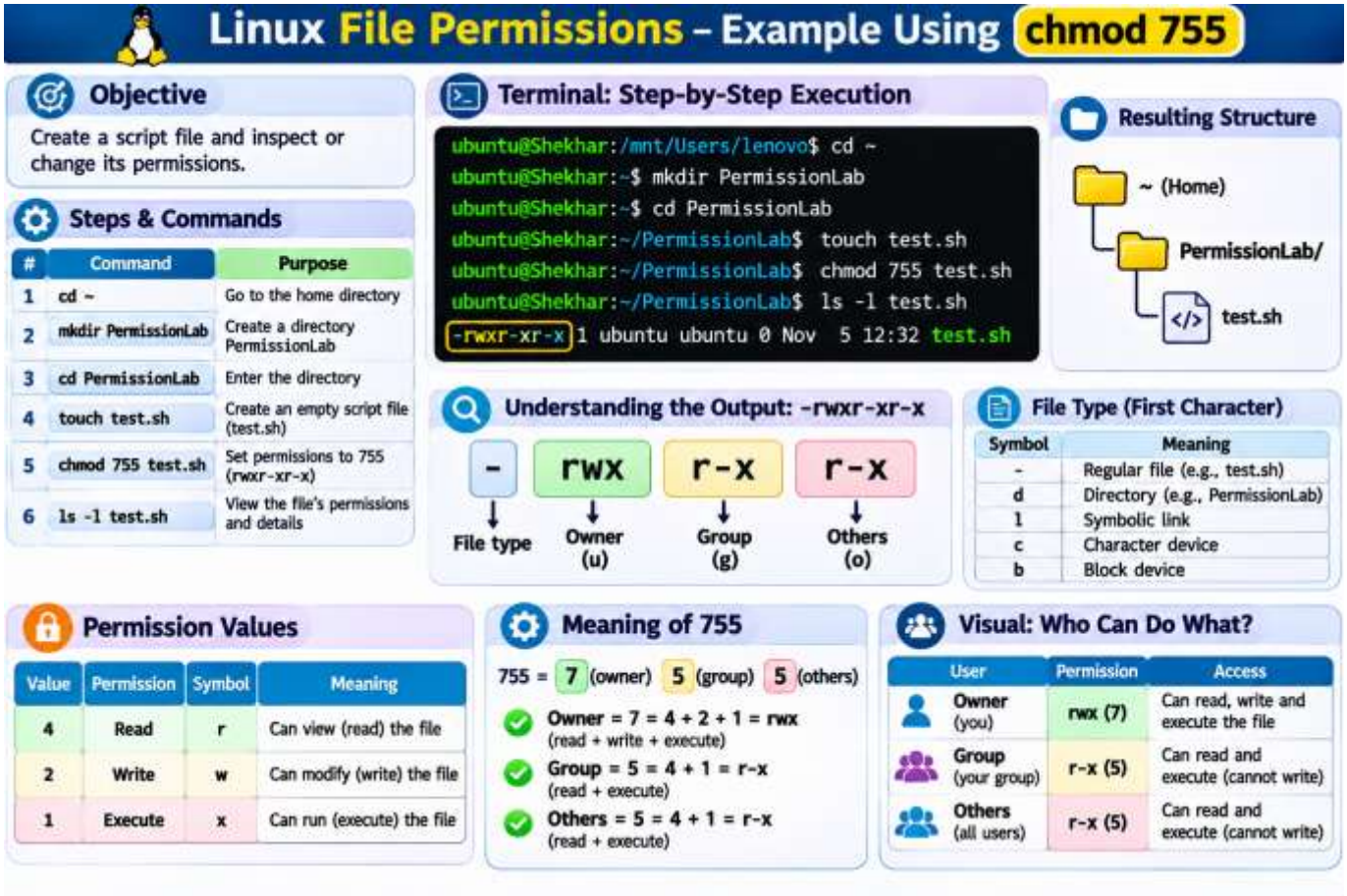
ubuntu@Shekhar:~$ **cd PermissionLab**

ubuntu@Shekhar:~/PermissionLab$ **touch test.sh**

ubuntu@Shekhar:~/PermissionLab$ **chmod 755 test.sh**

ubuntu@Shekhar:~/PermissionLab$ **ls -l test.sh**

-rwxr-xr-x 1 ubuntu ubuntu 0 Nov 5 12:32 test.sh



8. Process Management

Command	Purpose
ps	Display processes associated with the terminal
ps aux	Display a broad list of running processes
top	Live process and resource monitor; press q to quit
htop	Interactive process monitor (install if needed)
kill PID	Send the default termination signal to a process
kill -9 PID	Forcefully terminate a process; use only if normal termination fails
jobs	List jobs started from the current shell
bg	Continue a suspended job in the background
fg	Bring a background job to the foreground

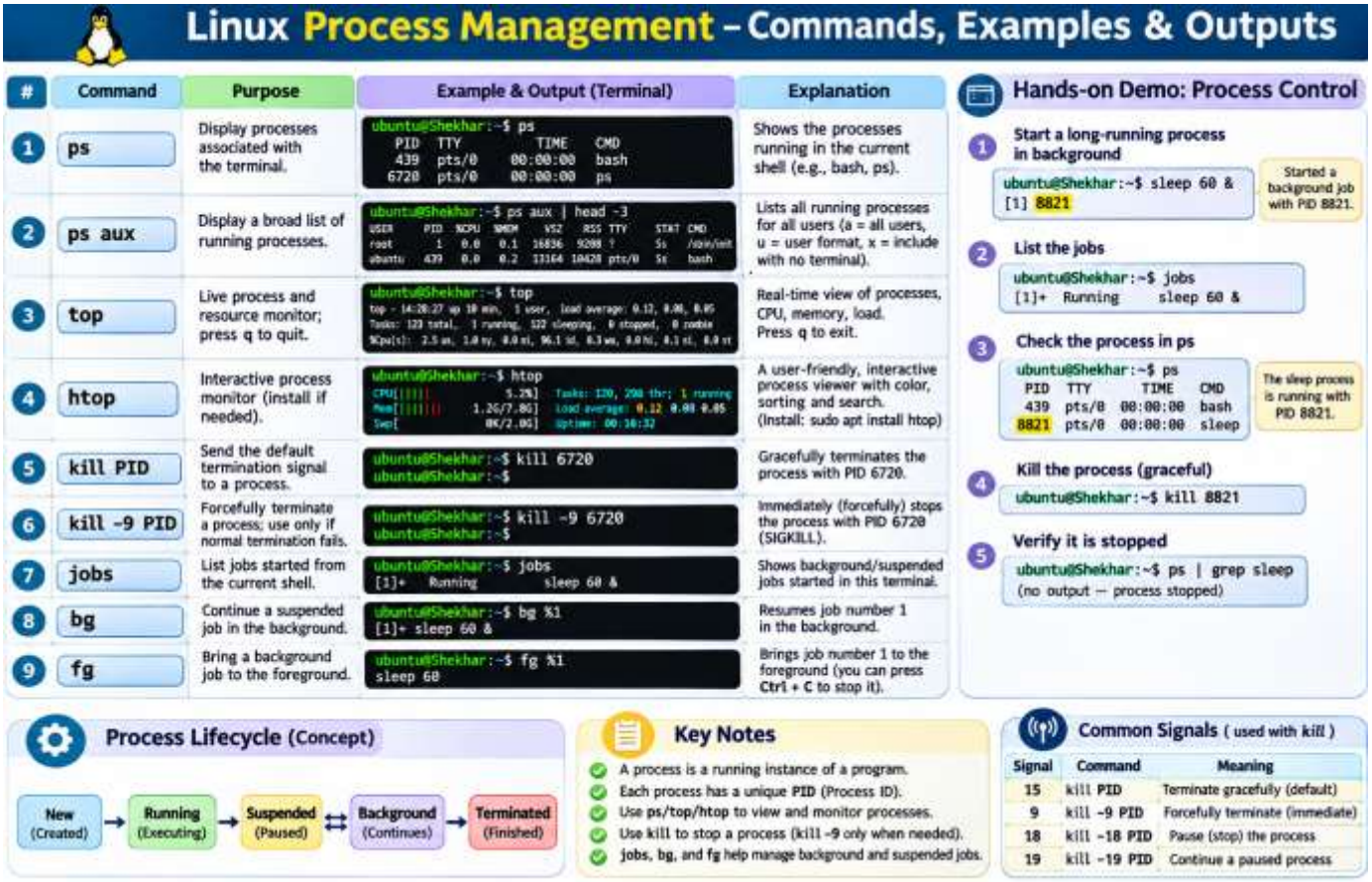
ubuntu@Shekhar:~/PermissionLab$ **cd ..**

ubuntu@Shekhar:~$ **ps**

PID TTY TIME CMD

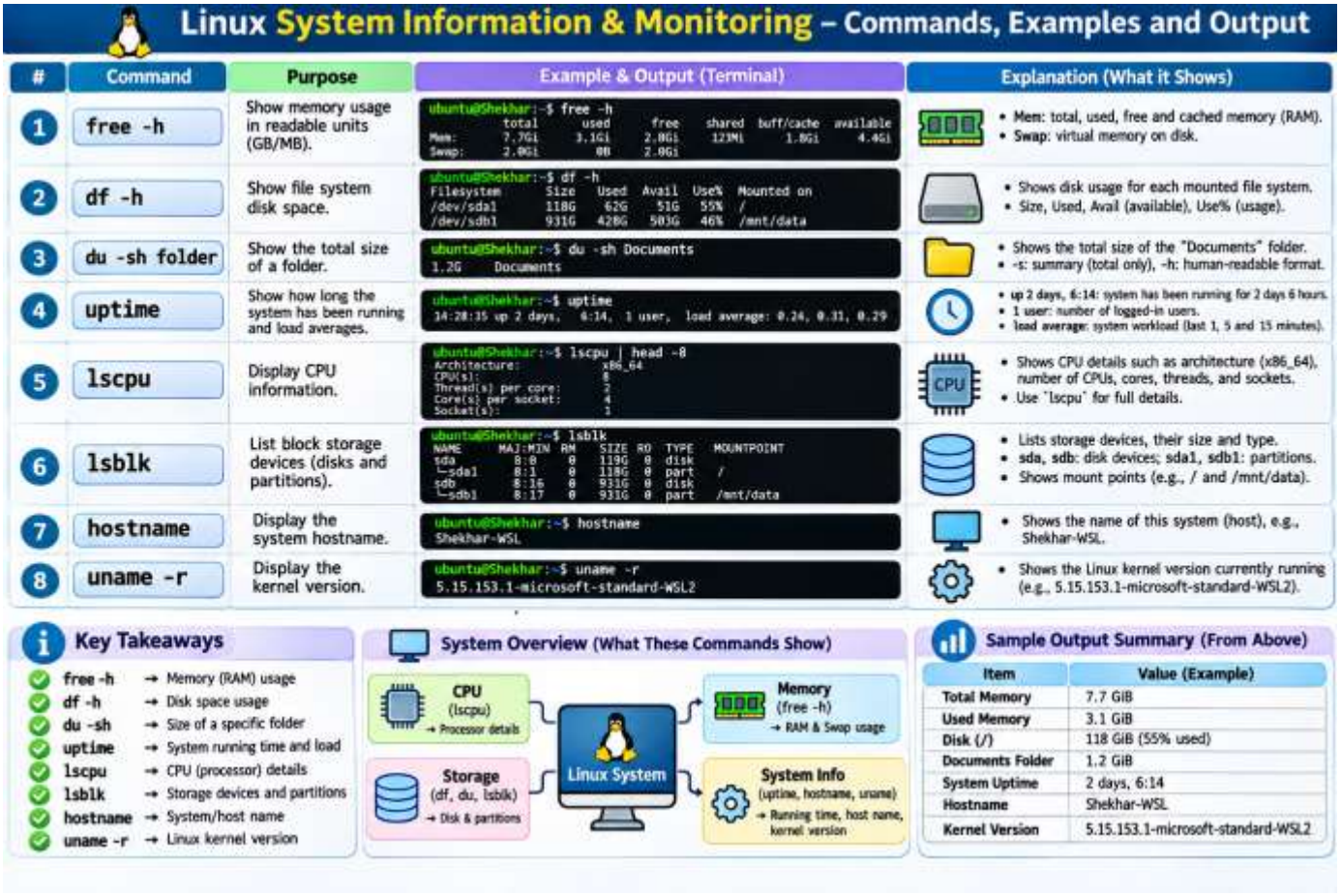
439 pts/0 00:00:00 bash

6720 pts/0 00:00:00 psReplace PIDwith the actual process ID shown by ps or top. Do not type the letters PID literally.



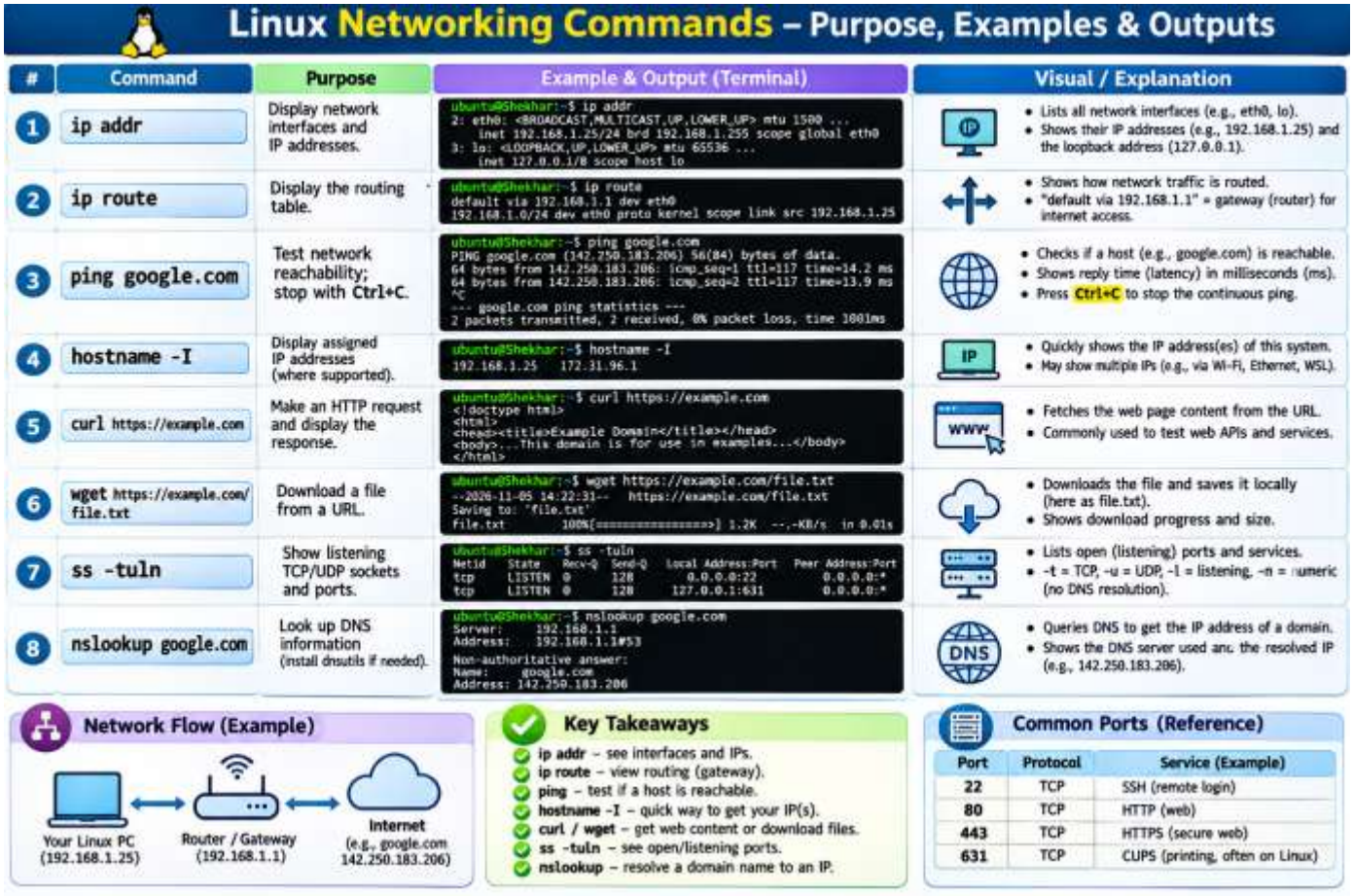
9. System Monitoring Commands

Command	Purpose
free -h	Show memory usage in readable units
df -h	Show file system disk space
du -sh folder	Show the total size of a folder
uptime	Show how long the system has been running and load averages
lscpu	Display CPU information
lsblk	List block storage devices
hostname	Display the system hostname
uname -r	Display the kernel version



10. Networking Commands

Command	Purpose
ip addr	Display network interfaces and IP addresses
ip route	Display the routing table
ping google.com	Test network reachability; stop with Ctrl+C
hostname -I	Display assigned IP addresses (where supported)
curl https://example.com	Make an HTTP request and display the response
wget URL	Download a file from a URL
ss -tuln	Show listening TCP/UDP sockets and ports
nslookup google.com	Look up DNS information (install dnsutils if needed)



11. Package Management in Ubuntu

APT is Ubuntu's package manager. Use it to refresh package information and install or remove software.

sudo apt

update sudo

apt upgrade

sudo apt install

tree sudo apt

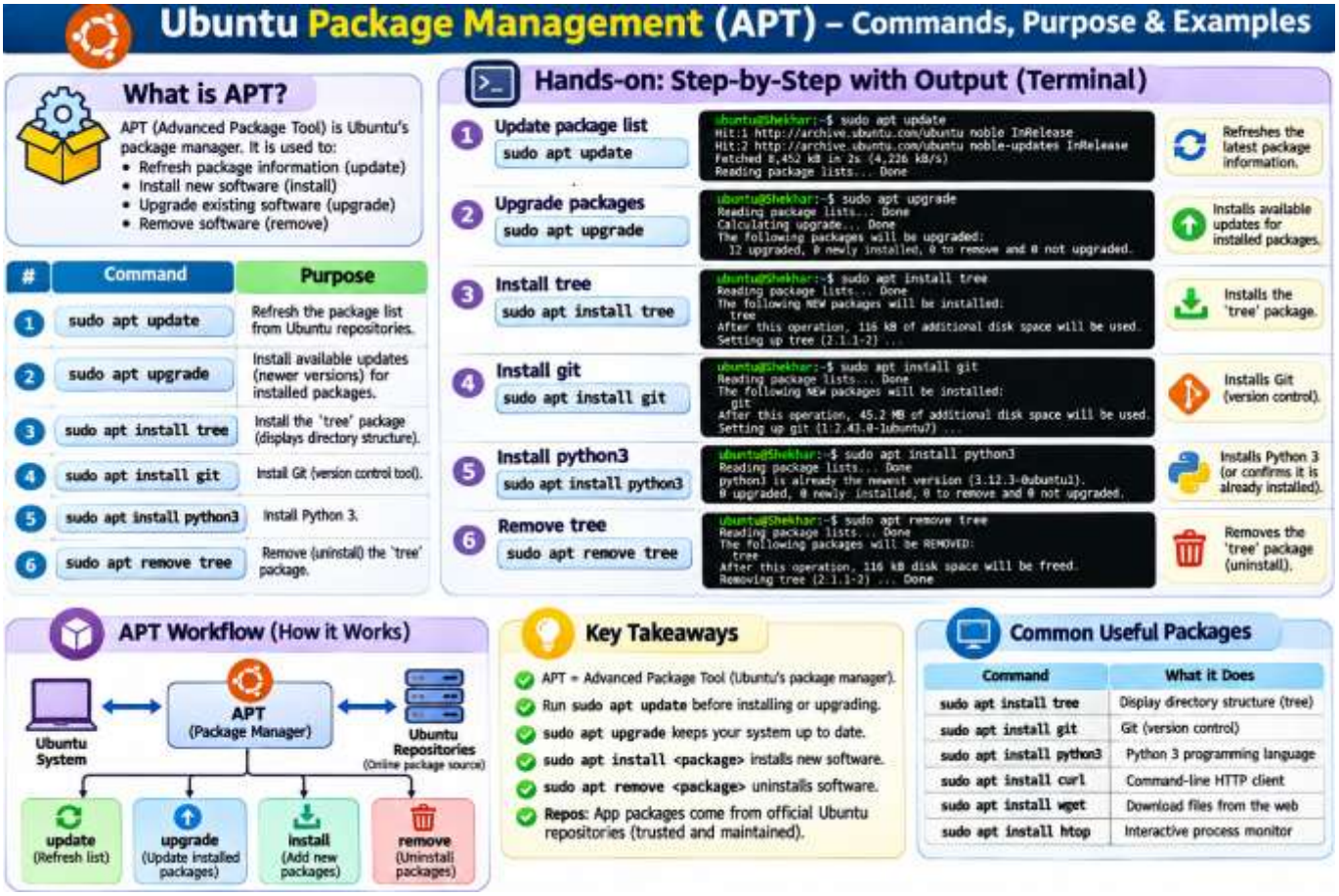
install git sudo apt

install python3

sudo apt remove

tree

The commands may ask for your Ubuntu password. Review the packages listed before confirming an upgrade or removal.

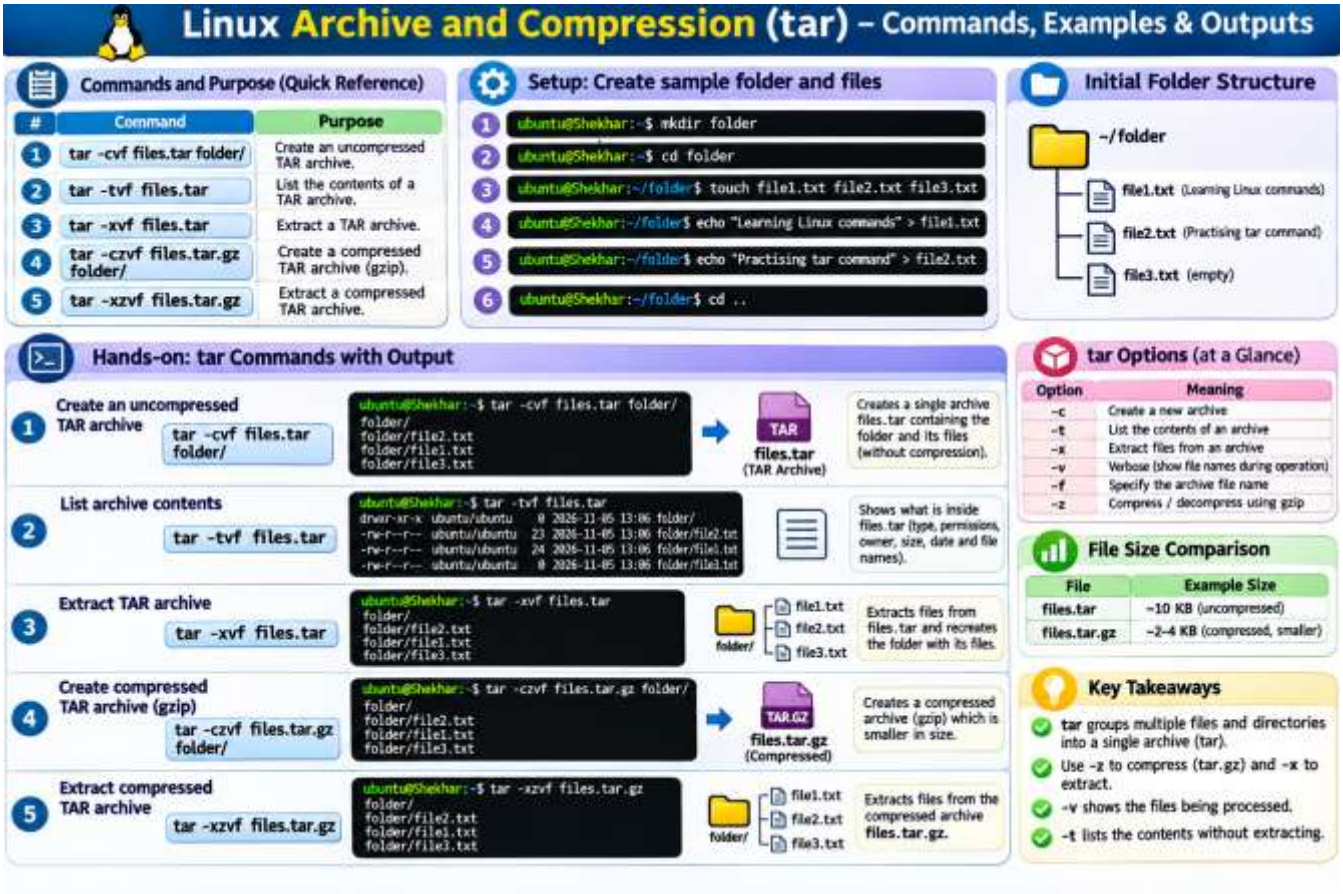


12. Archive and Compression

Command	Purpose
tar -cvf files.tar folder/	Create an uncompressed TAR archive
tar -tvf files.tar	Lists archive contents
tar -xvf files.tar	Extracts TAR archive
tar -czvf files.tar.gz folder/	Creates compressed TAR archive
tar -xzvf files.tar.gz	Extracts compressed TAR archive

ubuntu@Shekhar:~$ mkdir folder
ubuntu@Shekhar:~$ cd folder
ubuntu@Shekhar:~/folder$ touch file1.txt file2.txt file3.txt
ubuntu@Shekhar:~/folder$ echo "Learning Linux commands" > file1.txt
ubuntu@Shekhar:~/folder$ echo "Practising tar command" > file2.txt
ubuntu@Shekhar:~/folder$ cd ..
ubuntu@Shekhar:~$ tar -cvf files.tar folder/

```
folder/
folder/file2.txt
folder/file1.txt
folder/file3.txt
ubuntu@Shekhar:~$ tar -tvf files.tar
drwxr-xr-x ubuntu/ubuntu     0 2026-11-05 13:06 folder/
-rw-r--r-- ubuntu/ubuntu    23 2026-11-05 13:06 folder/file2.txt
-rw-r--r-- ubuntu/ubuntu    24 2026-11-05 13:06 folder/file1.txt
-rw-r--r-- ubuntu/ubuntu     0 2026-11-05 13:06 folder/file3.txt
ubuntu@Shekhar:~$ tar -xvf files.tar
folder/
folder/file2.txt
folder/file1.txt
folder/file3.txt
ubuntu@Shekhar:~$ tar -czvf files.tar.gz folder/
folder/
folder/file2.txt
folder/file1.txt
folder/file3.txt
ubuntu@Shekhar:~$ tar -xzvf files.tar.gz
folder/
folder/file2.txt
folder/file1.txt
folder/file3.txt
ubuntu@Shekhar:~$
```



13. Quick Reference: Essential Navigation

Command	Meaning
pwd	Where am I?
ls -la	What files and hidden files are here?
cd directory	Enter a directory
cd ..	Go up one level
cd ~	Go to Linux home
cd /mnt/c	Go to the Windows C: drive from Ubuntu/WSL
touch file.txt	Create an empty file or update its timestamps
cat file.txt	Read a file
cat > file.txt	Write/overwrite a file from terminal input
cat >> file.txt	Append to a file from terminal input
Ctrl+D	Finish terminal input for cat > or cat >>
Ctrl+C	Interrupt a running command

Suggested learning sequence: Day 1 — terminal basics and directories; Day 2 — file handling and searching; Day 3 — users and permissions; Day 4 — processes and monitoring; Day 5 — networking; Day 6 — package management and archives; Day 7 — revision and practical lab.

Safety reminder: Be careful with deletion and permission commands, especially rm -r, rm -rf, and chmod. Check your current directory with pwd before modifying or deleting files.